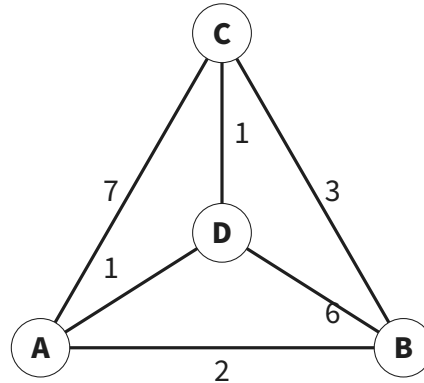


Cheapest pair first?

Name: _____

Date: _____

Start at A, travel every road at least once, return to A. Numbers are road lengths. Find the minimum total length. Repeated trips count again.



1. Roads and shortest connections differ.

What is the shortest connection from A to C? From B to D? A connection may use several roads.

2. Compare all three pairings.

All four vertices are odd. Pair them and repeat a shortest connection for each pair. Compare $AB+CD$, $AC+BD$, and $AD+BC$. Does beginning with the cheap pair AD guarantee the best total?

Connection costs, best pairing, and predicted total:

3. Certify the minimum.

Give an actual closed route attaining your total. For a lower bound, explain why the extra road copies in *any* closed solution can be separated into paths pairing the four odd vertices, plus possible closed loops.

Beyond four vertices: With $2k$ odd vertices, there are $(2k-1)(2k-3)\cdots 1$ pairings. Research on the Chinese postman problem replaces trying them all with efficient minimum-weight matching algorithms.

Try it with objects. Explain by pointing, talking, or drawing.